1. Identity

Document	STAR-PKG-STAR-LINSOLVE-EN · v2.2 · 2026-11-02
Package	star-linsolve
Version	0.1.0
Capability	star.numerics.linsolve
Maturity	Released & Verified (RELEASED)
Python	⩾3.9
Source commit	dc09f61
Source	https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_linsolve

2. Licence and obligations

Apache License 2.0. You may use, copy, modify and distribute it, also commercially, and you receive a patent licence from the contributors. When you redistribute it: keep the LICENSE file and the copyright notices (and a NOTICE file, if a future version adds one), and state in modified files that you changed them. You may not use the S.T.A.R. name to endorse your product.

3. What it needs at run time

Nothing beyond the Python standard library: it works offline.

4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_linsolve-0.1.0-py3-none-any.whl
```

FILE	SHA-256
star_linsolve-0.1.0-py3-none-any.whl	5b8db21a2a8f74058b78693eba2ce587760c0f4ac492640c926335658af0c4fb
star_linsolve.py	d48a71831dd3ba433b1eced9bdf0f1dc07557a69a3140515c53404bd6f93ead3
star_linsolve-0.1.0-source.zip	d40b9c98ccacb48720cd8a3773b5e22367affb9f3e6494cfcb55056d2d380407

5. What it does, and its limits

Determinant, solution and inverse of a square matrix of order 1 to 20 by elimination in exact rational arithmetic, each result rounded once: conditioning costs no digit and singularity is decided exactly; equal to SymPy's exact results on all 151 systems of the cross-check, Hilbert matrices of order 6 to 12 included.

Exact arithmetic on the floats you pass: 0.1 is not one tenth, and the solution of a badly conditioned system still changes a lot when the data change a little; the answer is exact for the matrix given, not for the one you meant. A determinant returned as 0.0 can be an underflow (two entries of 1e-200 on the diagonal) and is not by itself a proof of singularity: `solve` and `inverse` decide on the exact value. Not for large systems: nothing beyond order 20, no sparse matrices, no least squares, no eigenvalues, no condition number; a 20 × 20 inverse takes about a quarter of a second on a laptop and more when the entries have many different exponents.

- `det(a)` returns the determinant, `solve(a, b)` the vector x with a x = b, `inverse(a)` the inverse, for a matrix given as a list or tuple of rows.
- The elimination is done in exact rational arithmetic on the floats given and each result is rounded once: the conditioning of the matrix costs no digit. A matrix is singular exactly when its exact determinant is zero; no tolerance is involved, so a nearly singular matrix is solved and an exactly singular one is refused.
- Measured on 151 systems (110 well conditioned, 40 badly conditioned, and the Hilbert matrix of order 3 scaled to integers): determinant, solution and inverse equal the exact rational results of SymPy rounded to a float and those of mpmath at 400 digits. NumPy agrees within 1e-12 on the well-conditioned systems; on the badly conditioned ones it loses most digits or refuses the matrix as singular (the numbers are in the evidence file).

- Every function returns floats or raises `ValueError`: a matrix that is not square or larger than 20, entries that are not finite real numbers within 1e100 (booleans refused), a right-hand side of the wrong length, a singular matrix (for `solve` and `inverse`), a result too large for a float.

6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

7. Citation and bill of materials

```
@software{cavazzini_star_linsolve_0_1_0,
  author  = {Cavazzini, Andrea},
  title   = {{star-linsolve}: Small linear systems, correctly rounded},
  version = {0.1.0},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_linsolve/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit dc09f61}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

8. Warranty and liability

Provided "as is", without warranties or conditions of any kind (Apache License 2.0, sections 7 and 8), to the extent the law allows. This never excludes liability that the law does not allow to exclude.

Related: space activities and export control · acceptable use · evidence and limits · licence texts